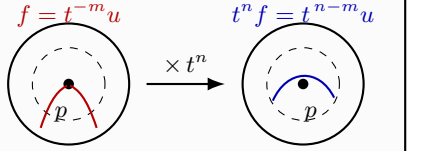
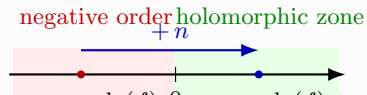
Local pole-clearing at one point



pole order m holomorphic iff $n - m \geq 0$

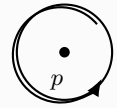
Multiply by the local correcting factor and ask that the singularity disappear.

Order language



$t^n f$ holomorphic at p
$\iff \mathrm{ord}_p(t^n f) \geq 0$
$\mathrm{ord}_p(t^n f) = \mathrm{ord}_p(f) + n$, so the condition becomes $\mathrm{ord}_p(f) + n \geq 0$.

Winding-form interpretation



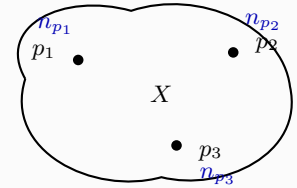
$$\mathrm{ord}_p(f) = \frac{1}{2\pi i}\int_{\gamma_p} \frac{df}{f}$$

$$n = \frac{1}{2\pi i}\int_{\gamma_p} n\,\frac{dt}{t}$$

$$\frac{1}{2\pi i}\int_{\gamma_p} \frac{d(t^n f)}{t^n f} = \mathrm{ord}_p(f) + n \geq 0$$

The divisor coefficient n is exactly the local winding added to cancel the negative winding of a pole.

Global compression into a divisor



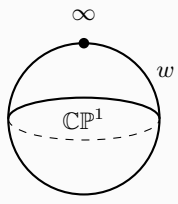
$D = \sum_p n_p[p]$, $\mathrm{ord}_p(f) + n_p \geq 0$ for all p

$\mathrm{div}(f) + D \geq 0$

$L(D) = \{f \in \mathcal{M}(X) \mid \mathrm{div}(f) + D \geq 0\} \cup \{0\}$

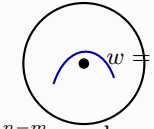
Because local holomorphicity is preserved under sums and scalar multiples, $L(D)$ is a vector space.

Model case on $\mathbb{CP}^1$: $D = n[\infty]$



$w = \frac{1}{z}$ near ∞

$f = z^m = w^{-m}$ $\times w^n$ $w^n f = w^{n-m}$ holomorphic iff $m \leq n$



w^{n-m} regular at 0

$$L\big(n[\infty]\big) = \mathrm{span}_{\mathbb{C}}\{1, z, z^2, \ldots, z^n\}$$

So the abstract inequality $\mathrm{div}(f) + D \geq 0$ is just the global form of the local rule: *prescribe correcting factors and require the corrected function to become holomorphic everywhere.*